

Manufacturing Junction gateKeeper AI Vision SOP

SOP-FAV-001 · Field installation, image monitoring, labeling, CPU fine-tuning and safe first operation · Revision 1.1.0

1. Purpose, scope and safety gate

Purpose

Inspect the FPCB model code before AAM input. The application classifies every image as **OK**, **Abnormal**, **Error** (registered problem code), or **System Error**.

Scope

Windows x64, CPU-only, one camera hot-folder, local labeling/training, sound alarm, audit archive and PLC **simulation/dry-run**.

First field use: PLC must remain Simulation or Dry-run. Do not map a physical stop, reject, or picker signal until the site safety review and independent real-lot validation are complete.

Required before start

- Windows 10/11 x64; 8 GB free disk minimum (16 GB RAM recommended for training).
- Camera software that saves one complete PNG/JPG per panel into one folder.
- Speaker connected if audible alarm is required.

Default recipe

- Normal expected code: **HJ04**
- Registered problem code: **HJ05**
- Only O/O is automatically equivalent; other character substitutions are not.

2. Install the offline ZIP and start the application

- Copy **Manufacturing-Junction-gateKeeper-AI-Vision.zip** to a local drive, for example **D:\gateKeeper**. Do not run from a network share or USB drive.
- Right-click → **Extract All...** → extract to a short local folder, for example **D:\gateKeeper\app**.
- Keep the entire extracted folder together. Do not move/delete **_internal**, **models**, **config**, **docs**, or either EXE file.
- Start **Manufacturing Junction gateKeeper AI Vision.exe**. If Windows asks for permission, obtain local IT approval before selecting **Run anyway**.

Pass condition: The dashboard opens without a dark-red System Error, shows HJ04/HJ05, and displays model status.

3. Connect image data and perform the safe three-image test



- Use the bundled **watch** folder or create, for example, **D:\gateKeeper\camera-watch**.
- Set the camera application's **Save image** destination to this exact folder. If possible, write a temporary extension first and rename to PNG/JPG only after writing completes.
- In gateKeeper Live Dashboard, select the same folder; set PLC to **Simulation/Dry-run**; press **Start monitoring**.

Live Dashboard (concept)

Expected normal: [HJ04 ▼] Problem codes: HJ05 PLC: Simulation
Folder: D:\gateKeeper\camera-watch

Manage code recipe **Start monitoring** **Mute** **Label images**

Put this known image into watch	Required result
Confirmed HJ04	Green OK ; no sound; archive/normal.
Confirmed HJ05	Red Error ; repeat alarm; simulated RejectRequest; archive/problem.
Blurred / unreadable / other code	Yellow Abnormal ; warning sound; never green; archive/abnormal or review.

Manufacturing Junction gateKeeper AI Vision SOP

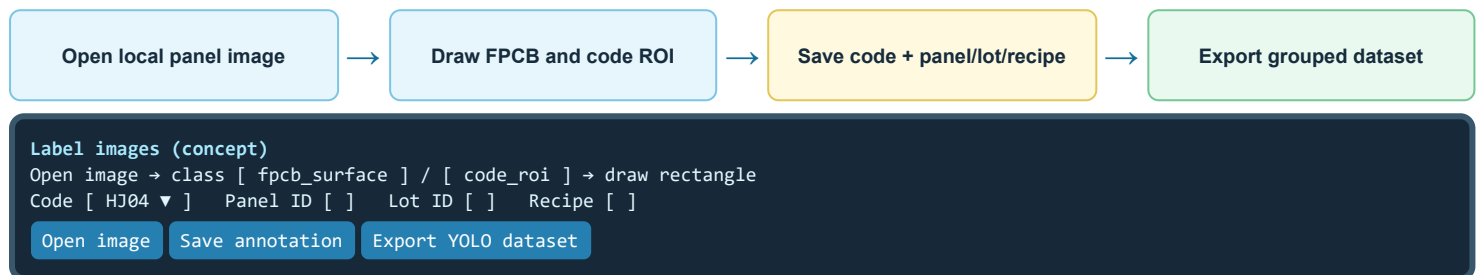
SOP-FAV-001 · Continue: recipe management, labeling, CPU fine-tuning and operating approval

4. Configure the code recipe

1. On the dashboard, press **Manage code recipe**.
2. Add/edit/delete only four-character uppercase letter/number codes. Keep normal and problem lists separate. Select the active expected normal code.
3. Press **Save**. The same recipe is used by live decisions, label validation and training labels.

Rule: HJ05 must remain in the problem list for the default test. An image with no ROI, OCR failure, or invalid length is **Abnormal**, never OK.

5. Label real local FPCB images



- 1 Press **Label images** → **Open image**. Use approved local field images only.
- 2 Select **fpcb_surface** and draw around the complete FPCB. Then select **code_roi** and draw tightly around only HJ04/HJ05.
- 3 Select the visible code; enter panel, lot and recipe/group; press **Save annotation**. Use the same lot ID for frames from one physical lot.
- 4 Collect varied focus, position, brightness, reflection and acceptable production variation. Use at least **three separate groups/lots**; do not split frames from the same panel/lot across train and test.

6. Export, CPU fine-tune, and decide whether to promote

1. Press **Export YOLO dataset**. Correct every label/split validation error; do not train with a warning unresolved.
2. Open **Training progress**. Use CPU, workers **0**, image size 640, batch **1–2** on low-memory PCs, 100 epochs, patience 20.
3. Start training. Review progress, validation graph, exact-code accuracy, problem recall, confusion matrix and saved best checkpoint.
4. Test the candidate only on a held-out real lot that was not used for labeling/training. Do not auto-replace the operating model.

Approve for production only when: held-out real-lot normal exact-code accuracy and problem recall meet the site target; no problem sample is classified as normal; ROI overlays are correct; and PLC fail-safe design is formally approved.

7. Live operation, feedback and recovery

Screen result	Operator action
OK	Confirm visual correctness during the trial. No alarm should remain.
Abnormal	Inspect panel/image. Use dashboard feedback: approve normal, correct to abnormal/problem, add to retraining data; enter operator and reason.
Error	Follow quality procedure. During this trial, confirm the simulated RejectRequest in audit logs only.
System Error	Stop monitoring. Check disk space, model files, camera folder permission and logs. Never treat it as normal.

End-of-shift checks

- ☐ HJ04 green, HJ05 red, blurred/unknown never green.
- ☐ Alarm/Mute, archive, overlay, duplicate guard and latest-popup replacement checked.

More detail in the app folder

docs/OPERATOR_GUIDE.md
docs/TRAINING.md
docs/SAFETY.md
docs/PLC_INTEGRATION.md

Stop and escalate: any System Error, yellow/red unexpectedly becoming green, incorrect ROI, missing archived image, or any proposal to enable physical PLC signals without safety approval.